

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
END SEM. THEORY EXAMINATION

Month and year: **Aug.'2026**

Program: **B.Tech.(CS, IT, AIML, AIDS, RA)**

Subject: **Programming Using Python**

Maximum Marks: **50**

Roll No. **125111044**

Total no. of pages used: **2**

Semester: **2nd (Regular/Reappear)**

Course code: **CSIC-104/ITIC-104/CSIC-114**

Time allowed: **03 Hours**

NOTE: 1. The question paper contains **FOUR** questions. 2. All Questions are compulsory.
3. Attempt all parts of a question **together at one place.**
4. While writing a Python program, **consider the following facts:**
(i) Follow the conventional and mandatory rules of Python programming.
(ii) Use appropriate names for all Identifiers. (iii) Make use of Indentation properly.
(iv) Show all commands for compiling/running your application.
(v) Show the entered input(s) & expected output(s).

Q1. Attempt **all parts** of the following:

[Marks: 2+2+2+2+2+4 = 14]

(i) What is the Python **REPL**? In an interactive REPL session, how can the result of the previous expression be used in the next expression/statement?

(ii) The **Python Virtual Machine (PVM)** manages multiple runtime memory areas during execution. What will be the content of the Method and Heap areas for the following Python Code snippet:

```
x = 10
def add(a, b):
    c = a + b
    return c
y = add(x, 20)
```

(iii) Write a Python program to receive two integer values from a user in a single call to the **input()**. Further, print the entered values separated by a tab space using a single call to the **print()** function.

(iv) Write a string slicing expression to extract the following portion (substring) of the string:

s = 'PYTHON'

(a) Last three characters

(b) Remove the last character

Note that you can only use negative index(s) while writing the slicing expression.

(v) Explain short-circuit (lazy) evaluation with a suitable example.

(vi) A local biologist needs a program to predict population growth. The inputs would be the initial number of organisms, the rate of growth (a real number greater than 0), the number of hours it takes to achieve this rate, and a number of hours during which the population grows. Write a program that takes these inputs and displays a prediction of the total population.

Q2. Attempt **all parts** of the following:

[Marks: 3 x 4 = 12]

(i) Which of the following expressions will make a single-element tuple: **x = (10)** or **x = (10,)**? With respect to the tuple data structure, state the truth value (True or False) of the following statements:

(a) A tuple can store different data types. (b) A tuple supports iteration. (c) A tuple supports slicing.

(ii) What are different built-in **set** types available in Python? Can we have a set as a member of another set? Justify your answer. Write a Python program that calculates the intersection of the sets **{1, 2, 3, 4}** and **{2, 4, 6}**.

(iii) Write a Python program that performs the following tasks sequentially:

- (a) Declare an empty dictionary. (b) Add two items (key/value pairs) using plain assignment to this empty dictionary; for example, {'Name': 'ram', 'Age': 23}
(c) Fetch all the values in the dictionary and iteratively print each value.

(iv) Create an empty list referenced by a variable named **myList**. Firstly, append the squares of the first five natural numbers to this empty list, one by one, using a for loop. Further, extend the list with a tuple having the first five prime numbers. Finally, remove the item at index -3 and print the final content of the list.

Q3. Attempt all parts of the following:

[Marks: 3 x 4 = 12]

(i) Design a class named **Demo** with a class attribute **x** which is initialized to the value 10. Further, define a class method and an instance method which simply print this class attribute **x**. Demonstrate the usage of your class by calling these methods.

(ii) Create a class **MyInt** which extends Python's built-in **int** class (an immutable type). When an object of your class is created by passing an integer value to its constructor, then the newly created object should have an integer value that is double the passed integer value. Demonstrate the usage of your class by creating an instance.

(iii) Design a class **Employee** with one instance attribute **salary**, which is initialized in the constructor using a parameter. Overload the **==** operator in such a manner that when we compare any two instances of the **Employee** class using the **==** operator, then it should return **True** if both objects have the same salary; otherwise, **False**. Write a Python program to illustrate the usage of your class.

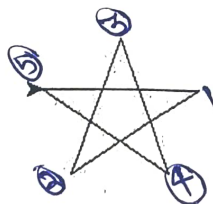
(iv) Write a Python program that defines a function using **variable-length arguments** (*args) to accept any number of integers and display their sum. Demonstrate the function by calling it with different numbers of arguments.

Q4. Attempt all parts of the following:

[Marks: 6 x 2 = 12]

(i) Write a GUI-based Python program using the **Tkinter** library that performs the addition of two integers entered by the user. The application should provide two text entry fields for accepting the input values and an "Add" button. When the user clicks the button, the program should compute the sum of the two integers and display the result in a label within the same window. The computed result should also be printed on the console.

- (ii)
(a) Write a Python program using the **Turtle Graphics** module to draw a **five-pointed star** as shown below.



(b) Write a Python program using the **Turtle Graphics** module to draw an expanding square spiral. The program should repeatedly move the turtle forward by an increasing distance, where the distance increases by 5 units in each iteration, starting from 0 units. After each forward movement, the turtle should turn 90 degrees to the right. The process should be repeated 50 times to produce the spiral pattern on the screen.